

CS4430 – Network Security Assignment 6: Network Telescope Analysis

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This report shows the results of analyzing network traffic from a telescope sensor. The goal is to find scanning behavior, top scanner IPs, and protocol use. The analysis was done using Python scripts on the provided PCAP files.

1 Top Scanners

Table 1 shows the ten most active scanning sources by packet volume. These IPs generated over 9% of total traffic, with 185.242.226.50 contributing 188,000+ packets, indicating automated scanning tools or botnet activity.

IP Address	Packets	Share
185.242.226.50	188,293	2.32%
185.242.226.46	105,263	1.30%
45.146.224.132	74,096	0.91%
10.14.176.3	72,407	0.89%
10.241.128.246	72,326	0.89%
80.75.212.37	59,683	0.74%
183.136.225.42	56,189	0.69%
212.70.149.138	54,256	0.67%
118.123.105.92	51,992	0.64%
218.92.0.99	49,420	0.61%

Table 1: Top 10 scanner IP addresses by packet count

2 Top Target Ports

Table 2 lists the top destination ports targeted by TCP and UDP scanners. Most TCP activity focused on ports commonly associated with remote access or web services (e.g., 22, 80, 443), while UDP scanning targeted DNS, SIP, and network service discovery. These results reflect broad automated probing for known services.

Port	Packets
22	174,938
3389	101,839
80	90,764
8088	82,746
8080	80,344
8728	76,159
5555	69,330
443	53,354
8122	51,620
12103	50,686

(a) TCP Scans

Port	Packets
514	144,733
53	91,129
5060	63,392
1194	40,973
32410	27,957
123	23,558
55001	13,926
1900	13,211
6881	13,009
5353	11,856

(b) UDP Scans

Table 2: Top 10 destination ports for TCP and UDP scans

3 Protocol Breakdown

Figure 1 shows the protocol breakdown of the captured traffic. TCP dominates with 7,404,342 packets (91.4%) versus UDP’s 697,351 packets (8.6%). This reflects typical scanning patterns targeting web and remote access services.

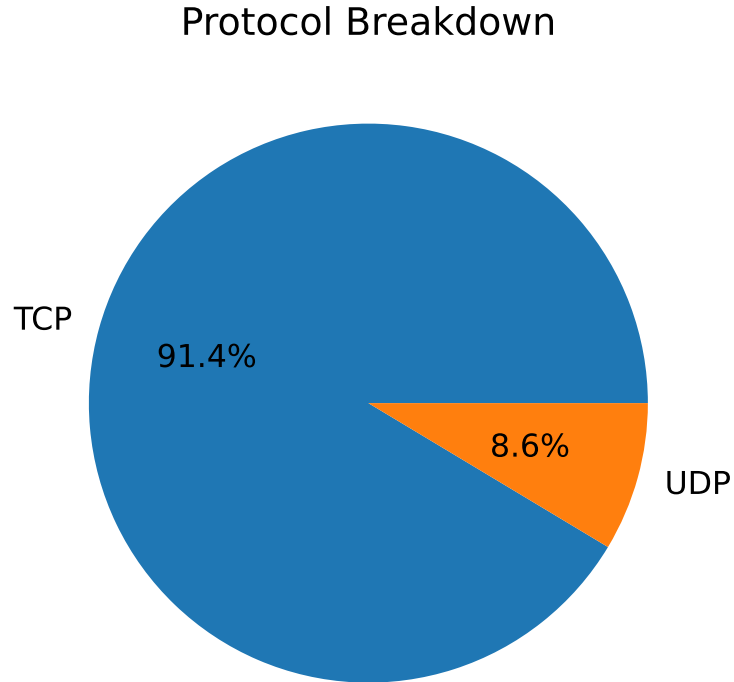


Figure 1: Protocol breakdown showing TCP vs UDP packet counts in the dataset

4 Heavy Hitter Scanner

Table 3 shows the scanner with the highest number of packets. IP address 185.242.226.50 sent over 188,000 TCP SYN and UDP packets, targeting 49,853 unique IP addresses and 4 different ports. This indicates a broad scanning strategy, likely automated. The IP geolocates to the Netherlands (country code NL).

Field	Value
Scanner IP	185.242.226.50
Number of packets	188,293
Number of destination IPs	49,853
Number of ports	4
Country	NL

Table 3: Profile of the heavy-hitter scanner

5 Traffic Visualizations

6 Interpretation and Reasoning